Goua Beedi

MSc MS2A | Machine Learning & AI Research

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Profile

After engineering studies in Statistics and Data Science at ENSAI, I am pursuing the MS2A master's degree at Sorbonne University. My experience has involved developing models, comparing their performance on real-world data and evaluating AI systems. I am now deepening my knowledge of probabilistic methods, simulation and generative models, with a focus on applied research.

Education

Sorbonne University Paris, France

MSc in Mathematics – Statistics, Learning and Algorithms (MS2A) *2026–2027*

Relevant coursework – Semester 1 (in progress): Statistical Learning, Stochastic Optimization, High-Dimensional Statistics, Data Science in Practice (machine learning, deep learning, generative models) and Simulation Methods (Monte Carlo, importance sampling, SMC, variational inference, VAEs, normalizing flows and diffusion models).

ENSAI – French National School for Statistics and Data Science Rennes, France

Engineering program in Statistics and Data Science *2022–2025*

Relevant coursework: Deep Learning (PyTorch), Statistical Learning at Scale, Natural Language Processing (NLP), Reinforcement Learning, Probability and Markov Models, Regularization, Numerical Optimization and Time Series.

Skills

Machine Learning & Deep Learning: Python, PyTorch, scikit-learn, XGBoost, neural networks, regression, classification, regularization and model selection.

Statistics & Modeling: linear regression and GLMs, probability, time series, statistical inference, calibration and optimization.

Simulation & Generative Models (coursework and ongoing project): Monte Carlo, importance sampling, SMC, variational inference, VAEs and guided diffusion models.

Research & Evaluation: benchmark design, group-based cross-validation, data leakage prevention, model comparison, error analysis and reproducible experimentation.

NLP & Engineering: RAG, embeddings, information retrieval, RAGAS, ChromaDB, NumPy, pandas, SQL, R, Git, Docker and FastAPI.

Experience

FINZUU – Fintech Paris, France

Data Scientist – Agricultural Credit Risk Scoring *2026*

- Developed Python-based scoring workflows using multi-source agricultural data to estimate production, income and repayment capacity.
- Built interpretable risk indicators, implemented consistency checks and integrated scoring workflows through FastAPI.

Square Management – Square Research Center Paris, France

AI & Data Science Intern – RAG System Evaluation *2025*

- Designed and executed a reproducible benchmark comparing **189 RAG configurations** on long documents, varying OCR, chunking, embeddings and retrieval methods.
- Evaluated retrieval relevance and answer faithfulness to source documents; analyzed performance differences and major failure modes.
- Automated evaluations with Python pipelines using RAGAS, ChromaDB, FastAPI and Docker.

City of Paris – Mobility Agency Paris, France

Data Science Intern – Mobility Data Analysis *2024*

- Combined heterogeneous LAPI, SIV and SIREN datasets, performed quality checks and developed statistical indicators for vehicle-fleet analysis and segmentation.

Projects

ENS Challenge – Parkinson's Disease

Challenge Data #159 – Individual Machine Learning Project, ongoing *2026*

- Modeled longitudinal patient trajectories using neural networks and Transformer architectures, with patient-disjoint cross-validation to prevent information leakage.
- Experimented with temporal corrections, compared and combined models, analyzed prediction errors and examined the consistency of predicted trajectories.

Diffusion Models & Rare Events – MS2A

Course Research Project – DireSMC, ongoing *2026–2027*

- Studying how diffusion models can be guided toward rare events using Sequential Monte Carlo (SMC), based on the paper *Steering Diffusion Models to Rare Events with Sequential Monte Carlo*. The project focuses on sampling methods and rare-event probability estimation.

Credit Scoring – Banque LCL

Academic Machine Learning Project *2025*

- Modeled default risk on **530,000 accounts and 143 features**; compared logistic regression, Random Forest and XGBoost, with feature selection, cross-validation and calibration analysis.

Languages

French: Fluent | **English:** Professional working proficiency